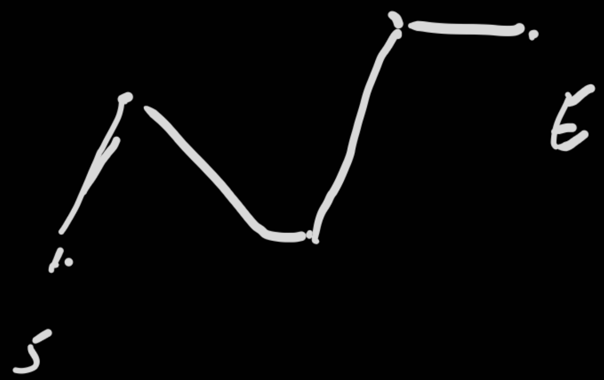
Vehicle node

Certificate of Participati

Route  (R)

Id of corresponding RSU

Position S (P) → Determines t
which the n

...on in VANET (C)

$\boxed{\text{RSU node}}$

$$R \longmapsto \sigma \cdot R : C$$

$$\sigma \in \mathrm{Aut}(\text{Traffic graph}) \qquad \sigma\, R' : C'$$

Public T...

...the RSU to

...nodes will connect

func collect : l = [ ] , local_map [[ ]] func

receive

No!

{ reh Id, R' } → append (l, R')

this is incorrect

should be hai

func aggregate :

Inbox
list

here

work on local_map → intention

key of

the encryption process is just

apply σ to the route. this encryption al

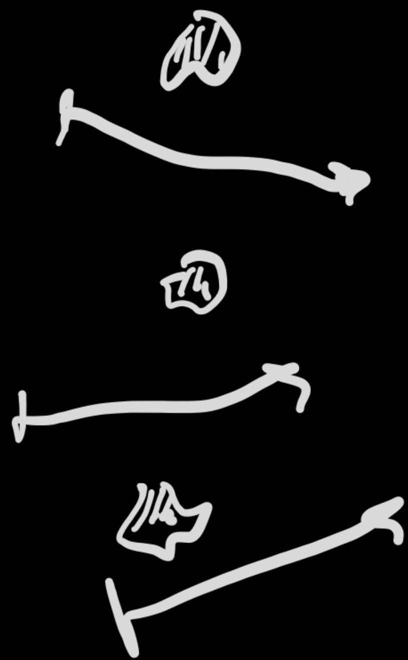
C

c report :

TMC ! aggregate (collect)

able,

lf



es

t to

g. is not secure. I just wanted a homomorphic operat

$\boxed{\text{TMC Node}}$

func assembl

from

assem

$\longrightarrow$

public_key $= \theta \in Aut(G)$

private_key $= \theta^{-1}$

road_capacities $:= \left\{ \begin{array}{l} (\,)- \\ (\,)- \\ (\,)- \\ \vdots \end{array} \right.$

u :

received real_world from RSU

ble a real_world map of the whole.

real_global_criph

urn.

fn decrypt :

obtain rea

$\longrightarrow$ real_

fn report :

(each road

of any ro

value encry

l_world_plain by applying α^{-1}

_world_plain

capacities to see if any overload

ad will happen. If so, return that

pText

(*) Now, since we

compute α^{-1}

messages form

Veh-

funct

ehicles have σ, they can also
thereby being able to decrypt
other vehicles

Node

danger:

receive

(Secure Channe
Model

In Erlang should be achie
hiding the Process Identifica

$\{ enc\_route, TMC \} = \{ d, \_ \} \rightarrow d \in \sigma CR$

Check if TMC classified any of its routes as ed)

ued by just

s, right?

)

dangerous.

+

Do we need a Router System f

not necessarily, right? If I let

communicate with themselves to

answer a vehicle its responsible

func move:

Just change its position,

changes the responsible

n this?

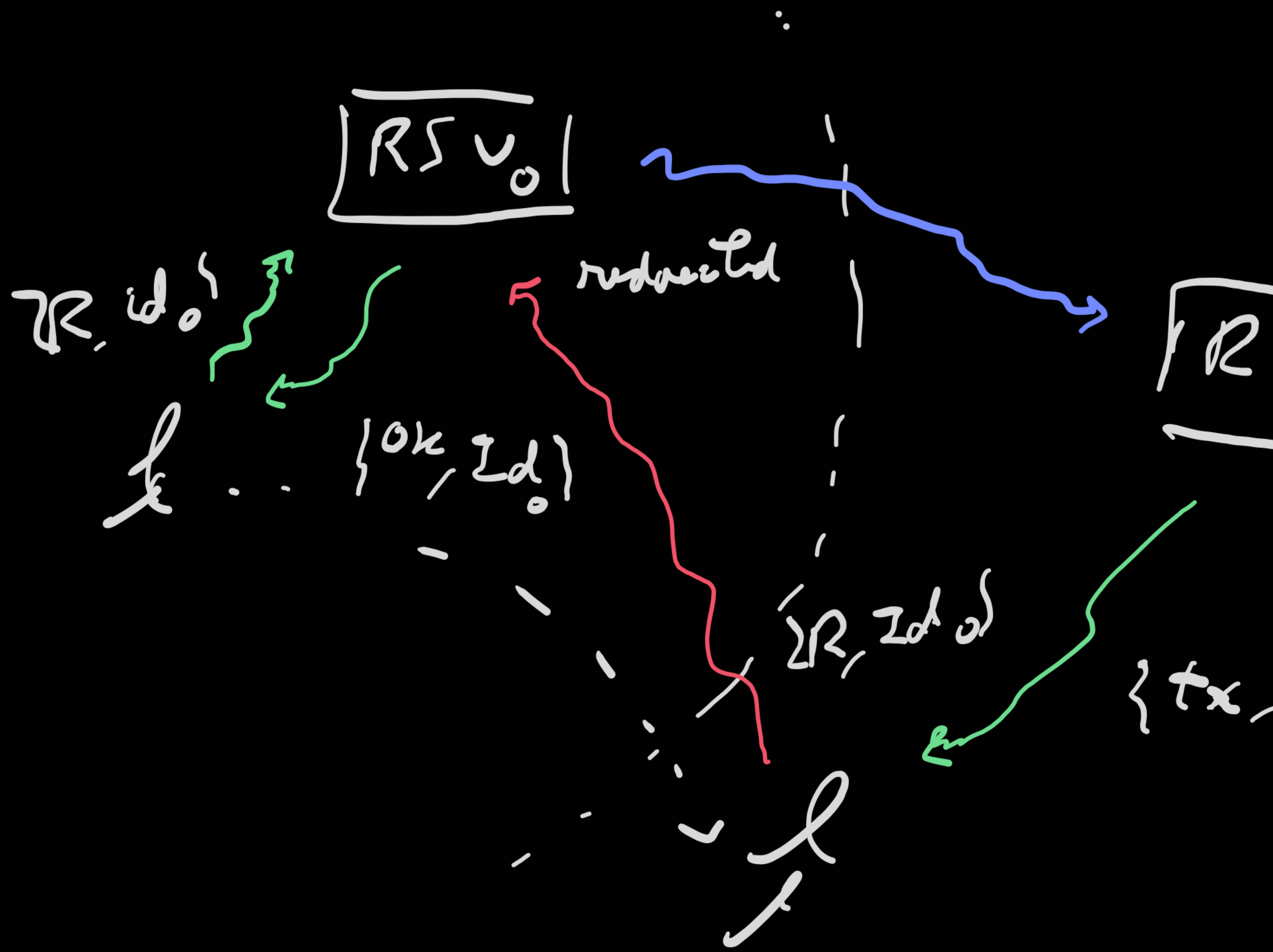
f the RSU

collected

RSU ..

which

RSU



$\boxed{SU_1}$

$+$

Key Server
node

$Id_1 \}$